1. Integral Terraces — Integration
    a. Integration e^x and 1/x

Example 1

Find $\int e^{2x+3}\,dx$.

$$\int e^{2x+3}\,dx = \frac{1}{2}(e^{2x+3})$$

The coefficient of x in $2x + 3$ will be a multiplying factor when the (e^{2x+3}) term is differentiated, so you need to include the $\frac{1}{2}$ to compensate for this.

When the integral has no limits (called an **indefinite integral**) as in Example 1, the integral is determined only up to a constant (the '$+c$' term). However, all the functions you will learn to integrate in this chapter can appear either as **definite integrals** (where you integrate between two limits), or as **improper integrals** (where the function is not defined at one end of the interval). Example 2 looks at a definite integral, and Example 3 at an improper integral.

Example 2

Calculate $\int_0^1 e^{2x+3}\,dx$.

$$\int_0^1 e^{2x+3}\,dx = \left[\frac{1}{2}e^{2x+3}\right]_0^1$$

The function is as in Example 1, but the $+c$ is not needed because we are integrating between two definite limits.

$$= \frac{1}{2}e^5 - \frac{1}{2}e^3$$

Evaluate the expression at the top and bottom limits.

$$= \frac{1}{2}(e^5 - e^3)$$

If we were to put the constant term $+c$ in the integral above, it would just cancel out when we evaluate the expression at the top and bottom limits. As such, we do not need to include the $+c$ when integrating any expression between two definite limits.

Example 3

Calculate $\int_{-\infty}^{1} e^{2x+3}\,dx$.

$$\int_{-\infty}^{1} e^{2x+3}\,dx = \left[\frac{1}{2}e^{2x+3}\right]_{-\infty}^{1}$$

The integral is as in Example 2, but the lower limit is different.

$$= \frac{1}{2}e^{5} - \lim_{a \to -\infty}\left(\frac{1}{2}e^{2a+3}\right)$$

Remember that (formally) this is the justification for saying the lower limit is 0.

$$= \frac{1}{2}e^{5} - 0$$

$$= \frac{1}{2}e^{5}$$

Example 4

Calculate $\int \sqrt{e^{x+1}}\,dx$.

$$\int \sqrt{e^{x+1}}\,dx \int \left(e^{x+1}\right)^{\frac{1}{2}}\,dx$$

The function needs to be expressed with a linear power.

$$= \int e^{\frac{1}{2}x+1}\,dx$$

The derivative of the power is $\frac{1}{2}$ so you need to multiply by 2 to compensate.

$$= 2e^{\frac{1}{2}(x+1)} + c$$

Example 5

Find the area bounded by the graph of $y = e^{2x}$, the x-axis, the y-axis, and the line $x = 1$.

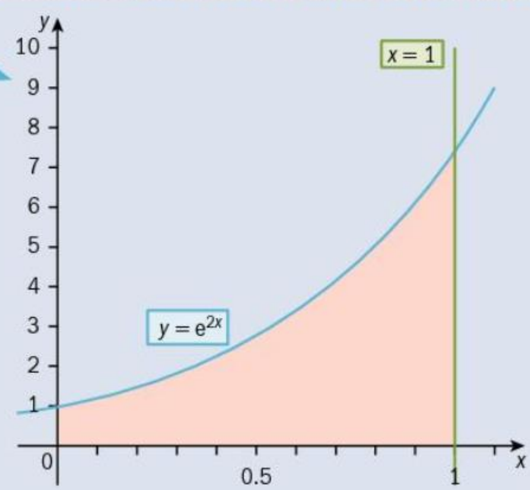
$$\text{Area} = \int_{0}^{1} e^{2x}\,dx$$

Sketching the graphs helps identify the required integral.

$$= \left[\frac{1}{2}e^{2x}\right]_{0}^{1}$$

Integrating and substituting the limits gives the area we're asked for.

$$= \frac{1}{2}e^{2} - \frac{1}{2}$$



Example 6

Find $\int \frac{2}{2x-3}\,dx$ and **state** the values of x for which the answer is valid.

$\int \frac{2}{2x-3}\,dx = \ln(2x-3) + c;$ valid for $x > \frac{3}{2}$

The derivative of the denominator is 2, so it is an exact integral, and $2x - 3 > 0$ when $x > \frac{3}{2}$.

Example 10

Calculate $\int_3^5 \frac{1}{3x-2}\,dx$, giving your answer as a single logarithm (i.e. in the form $\ln k$, where k is real).

$$\int_3^5 \frac{1}{3x-2}\,dx = \left[\frac{1}{3}\ln|3x-2|\right]_3^5$$

Use the standard form of the logarithmic integral.

$$= \frac{1}{3}\ln 13 - \frac{1}{3\ln}7$$

Evaluate the expression at the top and bottom limits.

$$= \frac{1}{3}\ln\left(\frac{13}{7}\right) = \ln\left(\sqrt[3]{\frac{13}{7}}\right)$$

First combine the two logarithms, then use the laws of logarithms to take the $\frac{1}{3}$ into the logarithm as the cube root.

Example 11

Calculate $\int_{-2}^{1} \frac{1}{7-2x}\,dx$, giving your answer as a single logarithm (i.e. in the form $\ln k$, where k is real).

$$\int_{-2}^{1} \frac{1}{7-2x}\,dx = \left[-\frac{1}{2}\ln|7-2x|\right]_{-2}^{1}$$

Use the standard from of the logarithmic integral, noting a is negative.

$$= -\frac{1}{2}\ln 5 - \left(-\frac{1}{2}\ln 11\right)$$

Evaluate the expression at the top and bottom limits.

$$= \frac{1}{2}\ln\frac{11}{5} = \ln\sqrt{2.2}$$

First combine the two logarithms, then use the laws of logarithms to take the $\frac{1}{2}$ into the logarithm as the square root.

Example 12

Calculate $\int_4^6 \frac{1}{7-2x}\,dx$, giving your answer as a single logarithm (i.e. in the form $\ln k$, where k is real).

$$\int_4^6 \frac{1}{7-2x}\,dx = \left[-\frac{1}{2}\ln|7-2x|\right]_4^6$$

The integral is the same as in Example 11, but with different limits.

$$= -\frac{1}{2}\ln 5 - \left(-\frac{1}{2}\ln 1\right)$$

Evaluating the expression at the top and bottom limits now requires taking the absolute values of −5 and −1.

$$= \frac{1}{2}\ln\frac{1}{5} = \ln\sqrt{0.2}$$

As before, first combine the two logarithms, then use the laws of logarithms to take the $\frac{1}{2}$ into the logarithm as the square root.

When ln 1 occurs in calculations, it can either be replaced by 0, or you can combine it into a single logarithm using the standard laws.

b. Integral sin/cos/sec^2

Example 13

Find $\int \cos 2x \, dx$.

$$\int \cos 2x \, dx = \frac{1}{2}\sin 2x + c$$

Remember to divide by the coefficient of x (i.e. 2).

there will not be a limit to which an integral approaches as x tends to infinity, so there will be no improper integrals of this sort, but care does need to be taken with the discontinuities that occur with $\tan x$ and also with all the reciprocal trigonometric functions.

Example 14

Find $\int \sin\left(3x + \frac{\pi}{4}\right) dx$.

$$\int \sin\left(3x + \frac{\pi}{4}\right) dx = -\frac{1}{3}\cos\left(3x + \frac{\pi}{4}\right) + c$$

Divide by the coefficient of x (i.e. 3).

c. Trig identity integrals

Example 17

Find the integrals

a) $\int \cos^2 x \,\mathrm{d}x$ **b)** $\int 12 \sin^2 x \,\mathrm{d}x$.

a) $\int \cos^2 x \,\mathrm{d}x = \int \frac{1}{2}(1 + \cos 2x)\,\mathrm{d}x$

$= \frac{1}{2}x + \frac{1}{4}\sin 2x + c$

b) $\int 12 \sin^2 x \,\mathrm{d}x = \int 12 \times \frac{1}{2}(1 - \cos 2x)\,\mathrm{d}x$

$= 6x - 3\sin 2x + c$

Use the double angle formula to get an integrable form.

Remember to integrate the number part as well as the trigonometric function, and include the $+ c$.

Again the double angle formula transforms this into an integrable form.

Example 18

Calculate the exact value of $\int_0^{\frac{\pi}{6}} 4\tan^2\theta \,\mathrm{d}\theta$.

$$\int_0^{\frac{\pi}{6}} 4\tan^2\theta \,\mathrm{d}\theta = \int_0^{\frac{\pi}{6}} 4(\sec^2\theta - 1)\,\mathrm{d}\theta$$

The use of the identity makes this a function you know how to integrate.

$$= [4\tan\theta - 4\theta]_0^{\frac{\pi}{6}}$$

These are standard integrals.

$$= \left(\frac{4}{\sqrt{3}} - \frac{4\pi}{6}\right) - (0 - 0)$$

Evaluate the expression at the top and bottom limits explicitly to show your working.

$$= \frac{4}{\sqrt{3}} - \frac{2\pi}{3}$$

Example 19

a) By writing $2x = 3x - x$ and $4x = 3x + x$, show that $\sin 3x \cos x = \frac{1}{2}(\sin 2x + \sin 4x)$.

b) Hence calculate $\int_0^{\frac{\pi}{6}} \sin 3x \cos x \,\mathrm{d}x$.

a) $\sin 2x = \sin(3x - x) = \sin 3x \cos x - \cos 3x \sin x$

$\sin 4x = \sin(3x + x) = \sin 3x \cos x + \cos 3x \sin x$

Using the compound angle formulae for sin x gives functions that you know how to integrate.

Adding the two expressions

$\Rightarrow \sin 2x + \sin 4x = 2\sin 3x \cos x$

$\Rightarrow \sin 3x \cos x = \frac{1}{2}(\sin 2x + \sin 4x)$

b) $$\int_0^{\frac{\pi}{6}} \sin 3x \cos x \,\mathrm{d}x = \int_0^{\frac{\pi}{6}} \frac{1}{2}(\sin 2x + \sin 4x)\,\mathrm{d}x$$

Using part (**a**).

$$= \left[-\frac{1}{4}\cos 2x - \frac{1}{8}\cos 4x\right]_0^{\frac{\pi}{6}}$$

Integrate the standard forms.

$$= \left(-\frac{1}{4}\times\frac{1}{2} - \frac{1}{8}\times\left(-\frac{1}{2}\right)\right) - \left(-\frac{1}{4} - \frac{1}{8}\right)$$

$$= \frac{5}{16}$$

Example 20

a) Show that $(3 \sin x - \cos x)^2 = 5 - 3 \sin 2x - 4 \cos 2x$.

b) Hence calculate $\int_0^{\frac{\pi}{2}} (3 \sin x - \cos x)^2 \, dx$.

a) $(3 \sin x - \cos x)^2 = 9 \sin^2 x - 6 \sin x \cos x + \cos^2 x$ ← Expand the square.

$= 9 \times \frac{1}{2}(1 - \cos 2x) - 3 \sin 2x + \frac{1}{2}(1 + \cos 2x)$ ← Use the standard formulae.

$= 5 - 3 \sin 2x - 4 \cos 2x$ ← Simplify to the required form.

b) $\int_0^{\frac{\pi}{2}} (3 \sin x - \cos x)^2 \, dx = \int_0^{\frac{\pi}{2}} (5 - 3 \sin 2x - 4 \cos 2x) \, dx$ ← Use part (**a**).

$= \left[5x + \frac{3}{2} \cos 2x - 2 \sin 2x\right]_0^{\frac{\pi}{2}}$ ← Integrate the standard forms.

$= \left(\frac{5\pi}{2} - \frac{3}{2} - 0\right) - \left(0 + \frac{3}{2} - 0\right)$

Be careful not to assume the limit at 0 is simply '0'.

$= \frac{5\pi}{2} - 3$ ← Simplify the answer.

Example 21

a) Find $\frac{d}{dx}(\cos x + x \sin x)$.

b) Hence calculate $\int_0^{\frac{\pi}{2}} (x \cos x) \, dx$.

Remember 'hence' means that you are expected to use a previous result.

a) $\frac{d}{dx}(\cos x + x \sin x) = -\sin x + \sin x + x \cos x = x \cos x$ ← Use the product rule.

b) $\int_0^{\frac{\pi}{2}} (x \cos x) \, dx = \int_0^{\frac{\pi}{2}} \left(\frac{d}{dx}(\cos x + x \sin x)\right) dx$ ← Use part (**a**).

$= \left[\cos x + x \sin x\right]_0^{\frac{\pi}{2}}$ ← Integration is the reverse process to differentiation.

$= \left(0 + \frac{\pi}{2}\right) - (1 + 0)$ ← Substitute at upper and lower limits.

$= \frac{\pi}{2} - 1$

Example 22

a) Show that $\frac{\cot^2\theta}{1+\cot^2\theta} = \cos^2\theta$. **b)** Hence calculate $\int_0^{\frac{\pi}{3}}\left(\frac{\cot^2\theta}{1+\cot^2\theta}\right)\mathrm{d}\theta$.

a) $\frac{\cot^2\theta}{1+\cot^2\theta} = \frac{\cot^2\theta}{\operatorname{cosec}^2\theta}$

Use the standard identity.

$$= \frac{\left(\frac{\cos^2\theta}{\sin^2\theta}\right)}{\left(\frac{1}{\sin^2\theta}\right)}$$

Express in terms of sin and cos only.

$$= \cos^2\theta$$

b) $\int_0^{\frac{\pi}{3}}\left(\frac{\cot^2\theta}{1+\cot^2\theta}\right)\mathrm{d}\theta = \int_0^{\frac{\pi}{3}}\left(\cos^2\theta\right)\mathrm{d}\theta$

Use part **(a)**.

$$= \int_0^{\frac{\pi}{3}}\left(\frac{1+\cos 2\theta}{2}\right)\mathrm{d}\theta$$

Use the standard form to convert to an integrable function.

$$= \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\frac{\pi}{3}}$$

Integrate.

$$= \left(\frac{\pi}{6} + \frac{\sqrt{3}}{8}\right) - (0+0)$$

Substitute at upper and lower limits.

$$= \frac{\pi}{6} + \frac{\sqrt{3}}{8}$$

Example 23

a) Show that $4\sin^4\theta = \frac{3}{2} - 2\cos 2\theta + \frac{\cos 4\theta}{2}$. **b)** Hence calculate $\int_0^{\frac{\pi}{2}} 4\sin^4\theta\,\mathrm{d}\theta$.

a) $4\sin^4\theta = (2\sin^2\theta)^2$

$$= (1-\cos 2\theta)^2$$

$$= 1 - 2\cos 2\theta + \cos^2 2\theta$$

$$= 1 - 2\cos 2\theta + \left(\frac{1+\cos 4\theta}{2}\right)$$

$$= \frac{3}{2} - 2\cos 2\theta + \frac{\cos 4\theta}{2}$$

Using the double angle cosine formula twice to remove powers of cos θ and sin θ gives the required result. An alternative approach is to start with the right-hand side and work down to single angle expressions.

b) $\int_0^{\frac{\pi}{2}} 4\sin^4\theta\,\mathrm{d}\theta = \int_0^{\frac{\pi}{2}}\left(\frac{3}{2} - 2\cos 2\theta + \frac{\cos 4\theta}{2}\right)\mathrm{d}\theta$

Use part **(a)**.

$$= \left[\frac{3}{2}\theta - \sin 2\theta + \frac{\sin 4\theta}{8}\right]_0^{\frac{\pi}{2}}$$

Integration is the reverse process to differentiation.

$$= \left(\frac{3\pi}{4} - 0 + 0\right) - (0 - 0 + 0) = \frac{3\pi}{4}$$

Substitute at upper and lower limits.

d. Integration with partial fractions

Example 1

Find $\int \frac{x+3}{(x-2)(x+1)}\,dx$.

$$\frac{x+3}{(x-2)(x+1)} \equiv \frac{5}{3(x-2)} - \frac{2}{3(x+1)}$$

From Example 1 in Chapter 7

$$\text{So } \int \frac{x+3}{(x-2)(x+1)}\,dx = \int \left(\frac{5}{3(x-2)} - \frac{2}{3(x+1)}\right)dx$$

Two terms which are each integrable as logarithmic functions

$$= \frac{5}{3}\ln(x-2) - \frac{2}{3}\ln(x+1) + c$$

Example 2 deals with fractions which have repeated linear factors in the denominator.

Example 2

Find $\int \frac{3x^2+2}{x(x-1)^2}\,dx$.

$$\frac{3x^2+2}{x(x-1)^2} \equiv \frac{2}{x} + \frac{1}{x-1} + \frac{5}{(x-1)^2}$$

From Example 3 in Chapter 7

$$\int \frac{3x^2+2}{x(x-1)^2}\,dx = \int \left(\frac{2}{x} + \frac{1}{x-1} + \frac{5}{(x-1)^2}\right)dx$$

Three terms which are each integrable

$$= 2\ln x + \ln(x-1) - \frac{5}{(x-1)} + c$$

Integrate each term carefully, and include the constant of integration.

Example 3

Find $\int \frac{2x^2 - x + 6}{(x+1)(x^2+2)} dx$.

$$\frac{2x^2 - x + 6}{(x+1)(x^2+2)} \equiv \frac{3}{x+1} - \frac{x}{x^2+2}$$

From Example 5 in Chapter 7

$$\int \frac{2x^2 - x + 6}{(x+1)(x^2+2)} dx = \int \left(\frac{3}{x+1}\right) dx - \int \left(\frac{x}{x^2+2}\right) dx$$

The first of these you already know how to integrate and the second you will meet in section 8.3.

$$= 3\ln(x+1) - \int \left(\frac{x}{x^2+2}\right) dx$$

Example 4 deals with improper algebraic fractions.

Example 4

Find $\int \frac{x^2 - x + 5}{(x-1)(x+2)} dx$.

$$\frac{x^2 - x + 5}{(x-1)(x+2)} \equiv 1 + \frac{5}{3(x-1)} - \frac{11}{3(x+2)}$$

From Example 7 in Chapter 7

$$\int \frac{x^2 - x + 5}{(x-1)(x+2)} dx = \int \left(1 + \frac{5}{3(x-1)} - \frac{11}{3(x+2)}\right) dx$$

Three terms which are each integrable

$$= x + \frac{5}{3}\ln(x-1) - \frac{11}{3}\ln(x+2) + c$$

Example 5

By first expressing $\frac{1}{(x-2)(x-3)}$ in partial fractions, find $\int \left(\frac{1}{(x-2)(x-3)}\right) dx$.

$$\frac{1}{(x-2)(x-3)} \equiv \frac{A}{x-2} + \frac{B}{x-3}$$

Or any standard technique to put into partial fractions

$$1 = A(x-3) + B(x-2); \quad x = 2 \Rightarrow A = -1; \quad x = 3 \Rightarrow B = 1$$

$$\int \left(\frac{1}{(x-2)(x-3)}\right) dx = \int \left(\frac{-1}{x-2} + \frac{1}{x-3}\right) dx$$

$$= -\ln(x-2) + \ln(x-3) + c$$

Integrate each term.

$$= \ln\left(k\left(\frac{x-3}{x-2}\right)\right)$$

Combine into a single logarithm.

Example 8

Find $\int\left(\frac{3x^3-4x^2-2}{x^2(x-1)}\right)\mathrm{d}x$.

$$\frac{3x^3-4x^2-2}{x^2(x-1)} \equiv A+\frac{B}{x}+\frac{C}{x^2}+\frac{D}{(x-1)}$$

There are now no improper fractions.

$$3x^3-4x^2-2 \equiv Ax^2(x-1)+Bx(x-1)+C(x-1)+Dx^2$$

Multiply each term by $x^2(x-1)$.

$$x=0 \Rightarrow -2=-C \Rightarrow C=2$$

$$x=1 \Rightarrow 3-4-2=-3=D \Rightarrow D=-3$$

$$3x^3-4x^2-2 = Ax^2(x-1)+Bx(x-1)+2(x-1)-3x^2$$

Substitute values for C and D.

$$= Ax^3-Ax^2+Bx^2-Bx+2x-2-3x^2$$

$$(x^3)\colon A=3;\ (x^2)\colon -4=-3+B-3 \Rightarrow B=2$$

Equate coefficients to find the last two constants; use the extra term to check.

Check coefficient of $x: 0=-B+2 \Rightarrow B=2$

$$\frac{3x^3-4x^2-2}{x^2(x-1)}=3+\frac{2}{x}+\frac{2}{x^2}-\frac{3}{(x-1)}$$

$$\int\left(\frac{3x^3-4x^2-2}{x^2(x-1)}\right)\mathrm{d}x=\int\left(3+\frac{2}{x}+\frac{2}{x^2}-\frac{3}{(x-1)}\right)\mathrm{d}x$$

$$=3x+2\ln x-\frac{2}{x}-3\ln(x-1)+c$$

Integrate each term.

$$=3x-\frac{2}{x}+\ln\left(\frac{x^2}{(x-1)^3}\right)+c$$

Combine the logarithmic terms.

Example 9

Evaluate $\int_4^6 \frac{1}{(x-2)(x-3)}\mathrm{d}x$.

$$\int_4^6 \frac{1}{(x-2)(x-3)}\mathrm{d}x=\left[-\ln(x-2)+\ln(x-3)\right]_4^6$$

Using Example 5

$$=(-\ln 4+\ln 3)-(-\ln 2+\ln 1)=\ln\frac{3}{2}$$

Combine into a single logarithm.

e. Integration of 1/1+x^2

Example 10

$y = \tan^{-1}\left(\frac{x}{a}\right)$. Find an expression for $\frac{dy}{dx}$.

$y = \tan^{-1}\left(\frac{x}{a}\right) \Rightarrow x = a\tan y$

$1 = a\sec^2 y\frac{dy}{dx}$

Differentiate with respect to x implicitly.

$1 = a(1 + \tan^2 y)\frac{dy}{dx}$

Use $1 + \tan^2 y = \sec^2 y$.

Rearranging gives

$\frac{1}{(1 + \tan^2 y)} = a\frac{dy}{dx}$

But $\frac{x}{a} = \tan y$, so

$\frac{1}{\left(1 + \left(\frac{x}{a}\right)^2\right)} = a\frac{dy}{dx}$

Substitute $\frac{x}{a}$ for $\tan y$ in $\frac{1}{1 + \tan^2 y}$.

Rearranging and simplifying:

$\frac{a^2}{a^2 + x^2} = a\frac{dy}{dx}$

$\frac{dy}{dx} = \frac{a}{a^2 + x^2}$

Example 12

Find $\int\left(\frac{1}{x^2+25}\right)\mathrm{d}x$.

$$\int\left(\frac{1}{x^2+25}\right)\mathrm{d}x=\frac{1}{5}\tan^{-1}\left(\frac{x}{5}\right)+c$$

Use $a=5$ in the standard result.

Example 13

Find $\int\frac{3}{x^2-8x+17}\mathrm{d}x$.

$$\int\frac{3}{x^2-8x+17}\mathrm{d}x=\int\left(\frac{3}{(x-4)^2+1}\right)\mathrm{d}x$$

Complete the square on the bottom.

$$=3\tan^{-1}(x-4)+c$$

Use the standard result.

Example 15

Find $\int \frac{1}{1+9x^2} \mathrm{d}x$.

$1+9x^2 = 1+(3x)^2$

$\int \frac{1}{1+9x^2} \mathrm{d}x = \frac{1}{3}\tan^{-1}(3x)+c$

Use the standard result.

Example 16

Find $\int \frac{1}{16+25x^2} \mathrm{d}x$.

$16+25x^2 = 16\left(1+\left(\frac{5}{4}x\right)^2\right)$

It is helpful to get the integrand into a standard form.

$\int \frac{1}{16+25x^2} \mathrm{d}x = \frac{1}{16}\int \frac{1}{1+\left(\frac{5}{4}x\right)^2} \mathrm{d}x = \frac{1}{16}\times\frac{4}{5}\tan^{-1}\left(\frac{5}{4}x\right)+c$

Use the standard result.

$= \frac{1}{20}\tan^{-1}\left(\frac{5}{4}x\right)+c$

Simplify the expression.
Remember to include $+c$ except where there are limits.

f. Integration by substitution

Example 17

Find $\int\left(\frac{2x}{x^2+2}\right)dx$.

$$\int\left(\frac{2x}{x^2+2}\right)dx = \ln(x^2+2)+c$$

Since $\frac{d}{dx}(x^2+2) = 2x$

Example 18

Find $\int\frac{2x^2-x+6}{(x+1)(x^2+2)}dx$.

$$\int\frac{2x^2-x+6}{(x+1)(x^2+2)}dx = 3\ln(x+1) - \int\left(\frac{x}{x^2+2}\right)dx$$

From Example 3

$$= 3\ln(x+1) - \frac{1}{2}\ln(x^2+2)+c$$

Using the result from Example 17

Example 19

Find $\int\left(\frac{e^{2x}}{e^{2x}+3}\right)dx$.

$$\int\left(\frac{e^{2x}}{e^{2x}+3}\right)dx = \frac{1}{2}\ln(e^{2x}+3)+c$$

Since $\frac{d}{dx}(e^{2x}+3) = 2e^{2x}$

Example 20

Find $\int \left(\frac{1 + 2x e^{x^2}}{x + e^{x^2}} \right) dx$.

$$\int \left(\frac{1 + 2x e^{x^2}}{x + e^{x^2}} \right) dx = \ln(x + e^{x^2}) + c$$

Since $\frac{d}{dx}(x + e^{x^2}) = 1 + 2x e^{x^2}$

Example 21

Find $\int \tan 2x \, dx$.

$$\int \tan 2x \, dx = \int \left(\frac{\sin 2x}{\cos 2x} \right) dx$$

Writing tan in this form allows you to view it as a logarithmic integral, since $\frac{d}{dx}(\cos 2x) = -2 \sin 2x$.

$$= -\frac{1}{2} \ln(\cos 2x) + c$$

$$= \frac{1}{2} \ln(\sec 2x) + c$$

This is an optional simplification.

Example 22

Find $\int_0^1 \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) dx$.

$$\int_0^1 \left(\frac{e^x - e^{-x}}{e^x + e^{-x}} \right) dx = \left[\ln(e^x + e^{-x}) \right]_0^1$$

Since $\frac{d}{dx}(e^x + e^{-x}) = e^x - e^{-x}$

$$= (\ln(e + e^{-1})) - (\ln(1 + 1))$$

Substitute the upper and lower limits.

$$= \ln(e + e^{-1}) - \ln 2$$

$$= \ln\left(\frac{e^2 + 1}{2e} \right)$$

Simplify the logarithmic function.

Example 32

Find $\int \sqrt{4-x^2}\,\mathrm{d}x$ using the substitution $x = 2\sin\theta$.

$$x = 2\sin\theta \Rightarrow \sqrt{4-x^2} = \sqrt{4-4\sin^2\theta} = 2\cos\theta$$

Only one expression here

$$1 = 2\cos\theta\frac{\mathrm{d}\theta}{\mathrm{d}x} \Rightarrow \mathrm{d}x = 2\cos\theta\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x$$

You need this to be able to complete the transformation of the integral to the new variable.

Substituting gives

$$\int\sqrt{4-x^2}\,\mathrm{d}x = \int 2\cos\theta \times 2\cos\theta\frac{\mathrm{d}\theta}{\mathrm{d}x}\mathrm{d}x$$

$$= \int 4\cos^2\theta\,\mathrm{d}\theta = \int 2(1+\cos 2\theta)\,\mathrm{d}\theta$$

Use the standard identity.

$$= 2\theta + \sin 2\theta + c = 2\theta + 2\sin\theta\cos\theta + c$$

Use the double angle relation.

$$= 2\sin^{-1}\left(\frac{1}{2}x\right) + \frac{1}{2}x\sqrt{4-x^2} + c$$

Express in terms of the original variable.

Example 33

Find $\int_1^2 x\sqrt{5x-2}\,dx$ using the substitution $u = 5x - 2$. Give your answer to 3 significant figures.

$u = 5x - 2 \Rightarrow x = \frac{u+2}{5}; \sqrt{5x-2} = u^{\frac{1}{2}}$ — The two terms in the integral

$x = 1 \Rightarrow u = 3; \quad x = 2 \Rightarrow u = 8$ — Change the limits.

$5 = \frac{du}{dx} \Rightarrow dx = \frac{1}{5}\frac{du}{dx}dx$ — So we can complete the transformation of the integral to the new variable

$\int_1^2 x\sqrt{5x-2}\,dx = \int_3^8 \left(\frac{u+2}{5}\right)u^{\frac{1}{2}}\,\frac{1}{5}\frac{du}{dx}\,dx$ — Substitute for the limits at the same time.

$= \frac{1}{25}\int_3^8 \left(u^{\frac{3}{2}} + 2u^{\frac{1}{2}}\right)du$ — Taking the common factor outside the integral simplifies things.

$= \frac{1}{25}\left[\frac{2}{5}u^{\frac{5}{2}} + \frac{4}{3}u^{\frac{3}{2}}\right]_3^8$ — These are standard integral forms now.

$= \frac{1}{25}\left(\frac{256}{5}\sqrt{2} + \frac{64}{3}\sqrt{2}\right) - \frac{1}{25}\left(\frac{18}{5}\sqrt{3} + 4\sqrt{3}\right)$

$= \frac{1}{25}\left(\frac{1088}{15}\sqrt{2} - \frac{38}{5}\sqrt{3}\right) = 3.5765... = 3.58\ (3\text{ s.f.})$

Because the limits are changed at the start, the definite integral does not need to be expressed in terms of x again.

Example 34

Find $\int_1^e \left(\frac{1}{x}\ln x\right)dx$ using the substitution $x = e^u$.

$x = e^u \Rightarrow \frac{1}{x} = e^{-u}; \quad \ln x = u$ — Express both parts in terms of u.

$x = 1 \Rightarrow u = 0; \quad x = e \Rightarrow u = 1$

$1 = e^u\frac{du}{dx} \Rightarrow dx = e^u\frac{du}{dx}\,dx$ — So we can complete the transformation of the integral to the new variable

Substituting gives — Substitute for all terms and the limits.

$$\int_1^e \left(\frac{1}{x}\ln x\right)dx = \int_0^1 e^{-u}\,ue^u\frac{du}{dx}\,dx$$

$$= \int_0^1 u\,du$$

Simplify the function and use the chain rule.

$$= \left[\frac{1}{2}u^2\right]_0^1$$

Integrate as a function of u.

$$= \frac{1}{2} - 0 = \frac{1}{2}$$

Evaluate the function at the limits.

Example 37

Find $\int \frac{1}{(5x-2)^2}\,dx$ using the substitution $u = 5x - 2$.

$u = 5x - 2 \Rightarrow \frac{1}{(5x-2)^2} = \frac{1}{u^2}$

As normal – express the function and dx in terms of u.

$5 = \frac{du}{dx} \Rightarrow dx = \frac{1}{5}\frac{du}{dx}\,dx$

$$\int \frac{1}{(5x-2)^2}\,dx = \int \left(\frac{1}{u^2}\right)\frac{1}{5}\frac{du}{dx}\,dx$$

$$= \frac{1}{5}\int\left(\frac{1}{u^2}\right)du$$

This is a standard integral form now.

$$= -\frac{1}{5u} + c$$

$$= -\frac{1}{5(5x-2)} + c$$

Express in terms of the original variable.

Writing $\frac{1}{(5x-2)^2}$ as $(5x-2)^{-2}$ allows this integral to be done without a formal substitution.

Example 38

Find $\int_0^{\frac{\pi}{4}} \cos 2x \sin^3 2x\,dx$ using the substitution $u = \sin 2x$.

$u = \sin 2x \Rightarrow \sin^3 2x = u^3$

Only the one expression here

$x = 0 \Rightarrow u = 0;\quad x = \frac{\pi}{4} \Rightarrow u = 1$

Change the limits.

$\frac{du}{dx} = 2\cos 2x \Rightarrow \frac{1}{2}\frac{du}{dx}\,dx = \cos 2x\,dx$

This is the second part of the function to be integrated.

$$\int_0^{\frac{\pi}{4}} \cos 2x \sin^3 2x\,dx = \int_0^1 \frac{1}{2}u^3\,du$$

Substitution makes the roles of the multiplying constants easier to see.

$$= \left[\frac{1}{8}u^4\right]_0^1$$

$$= \frac{1}{8} - 0 = \frac{1}{8}$$

Now the integration and evaluation are almost trivial.

g. Integration by parts

Example 23

Find $\int x\cos 3x \, dx$.

$$\int x\cos 3x \, dx = x\frac{1}{3}\sin 3x - \int\left(\frac{1}{3}\sin 3x\right)dx$$

$u = x$; $\frac{dv}{dx} = \cos 3x \Rightarrow v = \frac{1}{3}\sin 3x$

$$= \frac{1}{3}x\sin 3x + \frac{1}{9}\cos 3x + c$$

Don't forget the '$+\,c$' at the end here and be careful with negatives with the sin and cos integrals.

Example 24

Find $\int x^2 e^{2x} \, dx$.

$$\int x^2 e^{2x} \, dx = x^2\frac{1}{2}e^{2x} - \int\left(2x\frac{1}{2}e^{2x}\right)dx$$

$u = x^2$; $\frac{dv}{dx} = e^{2x} \Rightarrow v = \frac{1}{2}e^{2x}$

$$= \frac{1}{2}x^2 e^{2x} - \int(xe^{2x})\,dx$$

This needs integration by parts again.

$$= \frac{1}{2}x^2 e^{2x} - x\frac{1}{2}e^{2x} + \int\left(\frac{1}{2}e^{2x}\right)dx$$

Be careful about the negative signs when you use parts more than once.

$$= \frac{1}{2}x^2 e^{2x} - \frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + c$$

Example 25

Find $\int x^2\cos 2x \, dx$.

$$\int x^2\cos 2x \, dx = x^2\frac{1}{2}\sin 2x - \int\left(2x\frac{1}{2}\sin 2x\right)dx$$

$u = x^2$; $\frac{dv}{dx} = \cos 2x \Rightarrow v = \frac{1}{2}\sin 2x$

$$= \frac{1}{2}x^2\sin 2x - \int(x\sin 2x)\,dx$$

This needs integration by parts again.

$$= \frac{1}{2}x^2\sin 2x + x\frac{1}{2}\cos 2x - \int\left(\frac{1}{2}\cos 2x\right)dx$$

When integrating by parts more than once involving sin and cos, you need to be very careful with the negatives.

$$= \frac{1}{2}x^2\sin 2x + \frac{1}{2}x\cos 2x - \frac{1}{4}\sin 2x + c$$

Example 26

Find $\int (x \ln x)\,\mathrm{d}x$.

$$\int (x \ln x)\,\mathrm{d}x = \left(\frac{1}{2}x^2\right)\ln x - \int \left(\frac{1}{2}x^2\right)\left(\frac{1}{x}\right)\mathrm{d}x$$

$u = \ln x;\quad \frac{\mathrm{d}v}{\mathrm{d}x} = x \Rightarrow v = \frac{1}{2}x^2$

$$= \left(\frac{1}{2}x^2\right)\ln x - \int \left(\frac{1}{2}x\right)\mathrm{d}x$$

Now the second term on the right is recognisable as a standard integral.

$$= \left(\frac{1}{2}x^2\right)\ln x - \frac{1}{4}x^2 + c$$

Using the same process we can now find $\int (\ln x)\,\mathrm{d}x$ by thinking of $\ln x$ as the product $1 \times \ln x$.

It is not immediately obvious that writing in the form of a product will be helpful, but often in mathematics these little devices prove the key step in a useful result.

Example 27

Find $\int (\ln x)\,\mathrm{d}x$.

$$\int (\ln x)\,\mathrm{d}x = (x)\ln x - \int (x)\left(\frac{1}{x}\right)\mathrm{d}x$$

$u = \ln x;\quad \frac{\mathrm{d}v}{\mathrm{d}x} = 1 \Rightarrow v = x$

$$= x \ln x - \int (1)\,\mathrm{d}x$$

Again the second term on the right is recognisable as a standard integral.

$$= x \ln x - x + c$$